Fiscal Year:	FY 2023	**Task Last Updated:**	FY 08/22/2022
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metal Oxides		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
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PI Web Page:			
City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/21/2017	**End Date:**	05/20/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/20/2024 per NSSC information (Ed., 6/21/23) NOTE: End date changed to 05/20/2023 per NSSC information (Ed., 8/25/22) NOTE: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work (Ed., 5/18/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0059 ; NNX17AH73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. This project aims to measure the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential in order to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets is used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass provided the melts can be deeply supercooled. In some cases there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular viscosity and density) to advance models of the processing of materials. These models are important in advancing basic science of liquids and glasses and in development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at US universities and National Laboratories. ED. NOTE 5/18/2020: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is investigating molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. The project work includes outreach to undergraduate students. The experiments in microgravity are measuring essential thermophysical property data that can be used to advance understanding of how to process and optimize materials performance.
	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications, including functional glasses, batteries, next generation propulsion systems, lasers, and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. This project aims to measure the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential in order to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets are used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass, provided the melts can be deeply supercooled. In some cases there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular viscosity and density) to advance models for the processing of materials. These models are important in advancing basic science of liquids and glasses and in development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at U.S. universities and national laboratories. 1. Status: The project team includes Materials Development Inc. (MDI) scientists and Professor Robert Hyers, University of Massachusetts, Amherst; Dr. Shinji Kohara, National Institute for Materials Science, Japan; and Dr. Takehiko Ishikawa, JAXA, Japan. Experiments on ELF are performed by Dr. Chihiro Koyama and coordinated by Mr. Hirohisa Oda from JAXA. The collaborators from Japan are not funded from this grant. The grant was awarded on 5/21/2017 and assigned a new contract number in November 2017. The period of performance was extended. First year funding was at a level sufficient to scope the work and perform preliminary ground-based experiments in preparation for a more detailed experimental campaign during the subsequent years. Second year funding was at a level sufficient for planning and preparation for the Science Concept Review (SCR). The SCR was successfully completed in March 2020, and remaining issues from the SCR were addressed soon afterwards. The first round of flight experiments was started in November 2021 and completed in March 2022. This work included experiments on more than 30 samples. Due to some operational issues, a few samples could not be run. JAXA offered to provide a fourth sample holder, and MDI has prepared and supplied that holder’s additional samples. The plan is to launch the fourth holder on SpaceX CRS-27 (SpX-27) in January 2023. 2. Technical Activities: Activities during the fourth year included: (i) Run first set of flight experiments, (ii) continue analyzing data and recovered samples, (iii) make ground-based structure and property measurements on glass and liquid samples, and (iv) make additional samples for the fourth holder. Starting in August 2022, the MDI team is holding bi-weekly meetings with NASA Marshall Space Flight Center to discuss progress and plan future activities. (i) Flight experiments were run in ELF during the period November 2021-March 2022; five samples each of nine different compositions were launched. A precision stainless steel calibration sphere (identified as SUS) was measured at the start and end of each sample holder. As a result of ground-based tests, it was expected that the CA1, CA2, and CS1 samples would couple well with the ~980 nm wavelength heating laser used in ELF due to the addition of 2 mole % ytterbium oxide to the compositions. However, the laser absorption was insufficient to enable heating of the samples. After several unsuccessful attempts to melt these compositions, the remaining CA1, CA2, and CS1 samples were not run. An additional sample holder will be run in 2023 using CA1, CA2, and CS1

Task Progress:	compositions doped with iron oxide, which other JAXA investigators have used to successfully couple with the heating laser beam to enable full melting of the samples. The FS1 composition evaporated significantly during heating and full melting was not achieved. Initially all runs were performed using air as the gaseous medium. This worked well for the rare earth aluminate compositions. Density and drop oscillation measurements were performed on several samples of four different compositions as follows. YbAG – density from 1100-2450 ºC, no oscillation data, did not form glass; YbAP – density 400-2375, oscillation 1500-2100 ºC, did not form glass; YbLaAG – density 400-2250, oscillation 1535-2000 ºC, glass recovered; YbLaAP – density 4002500, oscillation 1980-2190 ºC, glass recovered. Analysis of the data for YbAG shows evidence of a potential density maximum in the supercooled liquid. This is being investigated further, including by making ground-based measurements of melt structure over the same temperature range in which density was measured. It was not possible to fully melt the mullite composition in air. Mullite samples were subsequently processed in argon. They could be fully melted in argon. Density was measured as a function of temperature from 650 to 2400 ºC. This included measurements on the deeply supercooled liquid. Drop oscillation measurements were made in the temperature range 1960 to 2200 ºC. Density exhibits a small temperature dependence. As described in our previous report, detailed structure measurements were made in ground-based experiments on the mullite composition. Since the aluminosilicates are of significant technological and geological importance, we have completed a detailed investigation of phase separation in the composition range from about 5-65 mole percent aluminum oxide. Two papers have been prepared and are currently in review. (ii) Data analysis is continuing. Data are being reduced using the initial sample mass in most cases. In some cases, sets of density data taken on different samples differ by several percent. We have not yet received all flight samples so the final mass that will be used for density analysis is not yet determined. Samples from the first holder were returned a few weeks ago and these can be analyzed with correct final masses. (iii) Ground-based tests are in progress to characterize the compositions for flight and to make measurements on the samples recovered from the flight experiments. Flight samples were examined by optical microscope to determine the overall appearance of the beads. High energy white beam X-ray tomography was performed at the Advanced Photon Source beamline 7 BM-B to determine the interior structure of the samples with a resolution of ca. 5 µm. These measurements are useful to evaluate the presence of bubbles or interior features that might affect the calculated volume (and hence density) of glass samples. In all cases, both flight and ground-based samples were examined in order to enable comparison of the materials processed in microgravity and on Earth. (iv) Samples for the fourth holder were supplied to JAXA in August 2022. The holder is scheduled for launch on SpX-27 in January 2023. Once the samples are on the International Space Station (ISS), detailed plans for the flight experiments will be finalized. The fourth holder will contain two pure aluminum oxide samples. These samples will be used to calibrate the temperature measurements using the well-established melting point of aluminum oxide as a reference. Two additional PRONTO samples will be launched with the fourth holder. The expected oscillation frequencies of molten samples were estimated using literature data for surface tension. Data were typically not available for the exact composition under study, so the values were obtained by extrapolating or interpolating data for similar materials from the literature [1-5]. The estimated oscillation frequencies provide a useful starting point for measurements in ELF. The initial coarse scan of frequencies used to obtain the resonant frequency of a liquid drop will be based on bracketing the estimated surface tension by +/- 100 mN/m. The typical frequency range is from about 120-160 Hz. 3. Publications/presentations: Papers were presented by Dr. Stephen Wilke and Dr. Richard Weber at the American Ceramic Society Glass and Optical Materials Division meeting, May 22-26, 2022 and the Physics of Non-crystalline Solids (PNCS-16) conference, July 10-15, 2022. Papers are being presented at the STOHT-2 meeting, Arizona State University, October 4-7, 2022 (Weber is an invited speaker) and the American Society for Gravitational and Space Research (ASGSR) annual meeting, Houston, November 9-12, 2022. Richard Weber was interviewed by Eileen DeGuire of the American Ceramic Society for a Ceramic Tech Chat podcast that was published in August 2022. An important topic of the interview is the flight experiments and their implications for glass development. https:// Two peer reviewed scientific papers were published: R. Weber, S.K. Wilke, and C.J. Benmore, “Containerless Techniques for in-situ X-Ray Measurements on Materials in Extreme Conditions,” J. Phys. Soc. Jpn., 91, 091008 (2022). G. Sivaraman, G. Csanyi, A. Vazquez-Mayagoitia, I.T. Foster, S.K. Wilke, R. Weber, and C.J. Benmore, “A Combined Machine Learning and High-Energy X-ray Diffraction Approach to Understanding Liquid and Amorphous Metal Oxides,” J. Phys. Soc. Jpn., 91, 091009 (2022). Both papers are open source. Two additional papers on aluminosilicates are in review. 4. Plans: Plans for the next project period are: (i) Continue data analysis, (ii) Continue sample analysis and characterization, (iii) Finalize plans for fourth holder ISS ELF experiments, (iv) Visit NASA Marshall Space Flight Center to perform ground-based experiments using the hybrid levitator, (v) Publish results. 5. References cited: [1] Kingery, J. Am. Ceram. Soc., 42, 6, (1959). [2] Bates, et al., Contract report NAS 3-14308, (1972). [3] Choi, et al., ISIJ International, 42, 221 (2002). [4] Weirauch, et al., J. Am. Ceram. Soc., 79, 920 (1996). [5] Conte, et al., Materials, 11, 2475 (2018).
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Weber R, Wilke SK, Benmore CJ. "Containerless techniques for in-situ X-ray measurements on materials in extreme conditions." J. Phys. Soc. Jpn. 2022 May 10;91:091008. https://doi.org/10.7566/JPSJ.91.091008 , May-2022

Articles in Peer-reviewed Journals	Sivaraman G, Csanyi G, Vazquez-Mayagoitia GA, Foster IT, Wilke SK, Weber R, Benmore CJ. "A combined machine learning and high-energy X-ray diffraction approach to understanding liquid and amorphous metal oxides." J. Phys. Soc. Jpn. 2022 May 10;91:091009. https://doi.org/10.7566/JPSJ.91.091009 , May-2022